

The Synthetic Economy: Can Small Language Models Generate Research-Grade Economic Microdata?*

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Abstract

Synthetic microdata are increasingly proposed as a privacy-preserving substitute for confidential survey data, yet most generative approaches require substantial compute and access to real training records. We investigate whether small, low-cost language models—Ling 3.0 Flash, DeepSeek V4 Flash, Solar Pro 4, and Qwen 3.7 Flash, priced at under \$0.03 each—can generate research-grade synthetic economic microdata from a text prompt alone, with no fine-tuning and no access to real records. Using public aggregated statistics from the U.S. Current Population Survey as ground truth, we generate 800 synthetic household records per model per condition (2,400 per model) across three prompt conditions (minimal, structural, priors) and evaluate them along four dimensions: statistical fidelity, downstream econometric utility, internal consistency, and diversity. We find that the models reproduce conditional economic structure remarkably well—education–income gradients within 4.7–11.6% mean absolute relative error, Mincer wage regressions with $R^2 = 0.86$ – 0.90 , and near-perfect internal consistency between employment, hours, and income—but systematically distort unconditional marginals, overstating the employment rate (62–83% vs. 63% reference) and the share of advanced degrees. A three-condition prompt ablation isolates the contribution of the models’ *latent* economic knowledge from *explicit* priors supplied in the prompt, revealing that explicit priors substantially improve income fidelity (MARE reduction of 40–53%) while having limited effect on econometric utility (R^2 changes < 0.04). Compared with two non-LLM baselines (uniform and marginal sampling), the LLMs’ unique value lies in their joint structure ($R^2 \geq 0.86$), not their marginals—a marginal sampler achieves MARE of 2.9% but R^2 of only 0.23. Cross-seed analysis demonstrates high reproducibility (coefficient of variation $< 3.4\%$ for R^2), and Wasserstein-distance analysis reveals that the synthetic income distributions are within 15–21% of the reference in normalized earth-mover distance. Our results establish that small language models can serve as cheap, zero-shot generators of structurally coherent economic microdata, while precisely identifying the marginals that require post-hoc correction before such data are fit for research use.

Keywords: synthetic data, large language models, economic microdata, Current Population Survey, Mincer regression, data fidelity, prompt ablation, reproducibility

*This work was produced with a total API budget of \$0.08 using four low-cost hosted language models. Code and data: <https://github.com/mnekrashevich/synthetic-economy>

1 Introduction

Confidential survey microdata—such as the U.S. Current Population Survey (CPS)—are the empirical backbone of labor economics, yet their release is tightly restricted to protect respondent privacy. Synthetic data generation has emerged as a leading strategy to reconcile the demand for granular microdata with privacy constraints [1]. Traditional generative approaches (GANs, VAEs, diffusion models) require training on the real data they aim to replace, which both demands substantial compute and inherits the very privacy risks they seek to avoid.

Large language models (LLMs) offer a qualitatively different route: because they encode broad world knowledge, they can in principle generate plausible microdata *zero-shot*, from a text description alone, without ever seeing a real record [2, 3]. This would make synthetic data generation dramatically cheaper and, by construction, free of direct record-level leakage. However, prior work has largely focused on large, expensive frontier models (e.g., GPT-4) and on fine-tuned or curated pipelines [1]. Whether a *small, cheap* model can produce research-grade economic microdata from a prompt alone remains an open and practically important question.

1.1 Motivation

The practical stakes are concrete. Researchers in developing countries often lack access to confidential microdata, yet must calibrate policy simulations against realistic population distributions. Educators teaching econometrics need realistic datasets for lab exercises but cannot distribute real CPS records. Data scientists prototyping ML pipelines for credit scoring or labor-market analysis need structurally coherent synthetic populations for unit testing. In each case, the desideratum is not perfect statistical fidelity, but *conditional structure*: the education–income gradient, the employment–hours–income consistency, and the concave age–earnings profile that makes a synthetic population behave like a real one in regression analysis.

This paper addresses that question with a deliberately minimal setup: four low-cost hosted models (Ling 3.0 Flash, DeepSeek V4 Flash, Solar Pro 4, and Qwen 3.7 Flash, each under \$0.03 for the entire study), a single text prompt encoding basic economic priors, and no fine-tuning, no real training data, and no curation. We generate 800 synthetic household records per model and benchmark them against public aggregated CPS statistics. Our contributions are:

1. A reproducible, near-zero-cost protocol for generating synthetic economic microdata from small LLMs, with total API spend under \$0.08 across four models.
2. A four-dimensional evaluation (fidelity, downstream utility, consistency, diversity) against public CPS aggregates, augmented with Wasserstein distances and cross-model Jensen–Shannon divergence, with bootstrap confidence intervals and two non-LLM baselines (uniform and marginal sampling) to contextualize the results.
3. A three-condition prompt ablation (minimal, structural, priors) across all four models that isolates the contribution of the models’ *latent* economic knowledge from the *explicit* priors supplied in the prompt.
4. Cross-seed reproducibility analysis demonstrating that the Mincer R^2 is stable across random seeds (coefficient of variation < 3.4%), establishing the reliability of the generation protocol.
5. Evidence that small LLMs capture *conditional* economic structure (education→income, employment→hours) far better than *unconditional* marginals (employment rate, education distribution), identifying precisely where post-hoc correction is needed.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes our generation and evaluation methodology. Section 4 presents results. Section 5 discusses implications and limitations. Section 7 concludes.

2 Related Work

2.1 Classical Synthetic Data Generation

Before LLMs, synthetic tabular data was generated by statistical and deep generative models. Copula-based methods, as implemented in the Synthetic Data Vault [4], model the joint distribution of features and sample from it. GAN-based approaches such as CTGAN [5] and its variational counterpart TVAE [6] learn to generate realistic rows from training data. In the economics domain, the **synthpop** package [7] synthesizes population microdata from real survey records. All of these methods share a common requirement: they must be *trained on the real data* they aim to replace, which both demands compute and inherits the privacy risks of the source data. Differential-privacy-aware generators [8] mitigate the latter but typically at a substantial cost to utility.

2.2 LLM-Based Tabular Data Generation

The use of language models to generate tabular data has grown rapidly. GReaT [2] established the core pipeline of serializing tabular rows as text and fine-tuning an LLM to generate new rows, demonstrating that language models can be realistic tabular data generators. Curated LLM [1] showed that in low-data regimes, a frozen LLM prompted with feature descriptions and a few examples can augment datasets, and that a principled curation mechanism substantially improves downstream utility. Fang et al. [3] survey the broader landscape of LLM methods for tabular prediction, generation, and understanding. More recent work has explored zero-shot generation from structured prompts [9], privacy-preserving fine-tuning [10], and the use of smaller, cheaper models for data augmentation [11]. Several 2025–2026 studies have specifically examined the fidelity of LLM-generated tabular data, finding that instruction-tuned models can produce realistic distributions when provided with detailed schema descriptions and marginal constraints [12, 13].

2.3 LLMs for Economic and Social Science Data

A smaller but growing literature applies LLMs specifically to economic data generation. Ante et al. [14] use LLMs to generate synthetic economic indicators for nowcasting. Huang et al. [15] survey the use of LLMs in social science research, noting their potential for generating training data. However, no prior study systematically evaluates whether small, low-cost LLMs can reproduce the *econometric structure* of a real household survey—specifically, whether a Mincer wage regression fit on synthetic data recovers canonical economic parameters.

2.4 Gap

Two features distinguish our work from this prior art. First, prior LLM generators either fine-tune the model [2] or provide in-context examples and curation [1]; we use a *pure zero-shot* prompt with no examples and no curation, isolating the model’s prior knowledge. Second, prior work emphasizes large frontier models; we deliberately use small, sub-\$0.03 models to test the practical floor of what is achievable. To our knowledge, no prior study evaluates whether small LLMs can reproduce the *econometric structure* of a real survey—specifically, whether a Mincer wage regression fit on synthetic data recovers canonical economic parameters—across multiple models under

controlled prompt ablation. Crucially, we also compare against simple non-LLM baselines (uniform and marginal sampling), which prior LLM tabular-generation papers rarely do, and we run a prompt ablation to separate latent knowledge from instructed priors. Finally, we provide cross-seed reproducibility analysis and Wasserstein-distance metrics to complement standard MARE and R^2 measures.

3 Methodology

3.1 Problem Formulation

We study the following problem: given a text prompt describing a population survey schema and (optionally) aggregated statistics, can a small language model generate a set of synthetic records $\tilde{X} = \{\tilde{x}_1, \dots, \tilde{x}_n\}$ whose statistical properties approximate those of a real population X ? We evaluate approximation along four dimensions: (i) marginal fidelity—how close are the synthetic univariate distributions to the reference?; (ii) conditional fidelity—how well do the synthetic data preserve relationships between variables?; (iii) downstream utility—can a researcher obtain similar econometric estimates from synthetic data as from real data?; and (iv) consistency—are the synthetic records internally logically coherent?

Our evaluation uses public aggregated statistics from the U.S. Current Population Survey [16, 17, 18] as the ground-truth reference X . We do not have access to the actual CPS microdata; this is by design, as our protocol must be reproducible without restricted data access.

3.2 Models

We evaluate four small, low-cost hosted language models served via the OpenRouter API: Ling 3.0 Flash (\$0.021/M input, \$0.063/M output), DeepSeek V4 Flash (\$0.030/M, \$0.160/M), Solar Pro 4 (\$0.030/M, \$0.120/M), and Qwen 3.7 Flash (\$0.025/M, \$0.080/M). All four models were released or updated in mid-2026 and represent the current state of small, affordable language models. Reasoning was disabled (“effort: none”) where supported to reduce cost and ensure clean JSON output. No real records, examples, or in-context demonstrations were provided. The total API cost across all four models and all conditions was under \$0.08.

3.3 Data Generation and Prompt Ablation

To separate the models’ *latent* economic knowledge from the *explicit* priors we supply, we run a three-condition prompt ablation:

- **Minimal:** the prompt specifies only the field schema (age, gender, education, income, employed, hours, marital status, children, state), with no economic guidance.
- **Structural:** adds logical consistency constraints—employed people work positive hours, the non-employed earn zero income, and income follows an age lifecycle—but no target values.
- **Priors:** additionally supplies explicit target values (median income by education, education distribution), matching the original protocol.

The full prompts are reproduced in Appendix A.

Each API call requested a JSON array of 20 records. We generated 400 records per seed across two seeds (42, 2024), yielding 800 records per model per condition. Each record contains nine fields: `age` (18–85), `gender`, `education` (high_school, some_college, bachelors, masters, phd),

`income` (annual USD), `employed` (boolean), `hours_worked` (0–80), `marital_status`, `children` (0–6), and `state` (two-letter US state). Records failing basic sanity checks (out-of-range values, invalid categories) were discarded. The total dataset comprises 4 models $\times$ 3 conditions $\times$ 800 records = 9,600 synthetic records.

3.4 Baselines

To contextualize the LLMs’ performance, we compare against two non-LLM generators that require no training:

- **Uniform:** each field is sampled independently from a uniform distribution over its support. This is the “no knowledge” floor.
- **Marginal:** education is sampled from the Census distribution, employment from the BLS unemployment-by-education rates, and income from a log-normal centred on the BLS median for the sampled education level. This is the “marginals only, no correlations” baseline.

These baselines isolate what is achievable from public marginals alone, without any joint structure.

3.5 Ground-Truth Reference

We benchmark against public aggregated statistics from the Current Population Survey, drawn from BLS and Census publications:

- **Median income by education** (BLS 2024, full-time workers 25+, annualized from weekly): high school \$48,360, some college \$53,040, bachelor’s \$80,236, master’s \$95,680, PhD \$118,456 [16].
- **Educational attainment** (Census 2024, adults 25+): high school or less 36.8%, some college 25.4%, bachelor’s 23.5%, master’s 11.0%, PhD 3.3% [18, 19].
- **Unemployment by education** (BLS 2024): high school 4.2%, some college 3.8%, bachelor’s 2.5%, master’s 2.2%, PhD 1.2% [17].
- **Employment-population ratio** (adults 25+, BLS 2024): approximately 63%.
- **Mincer return to schooling:** canonical estimate of 8–10% per year [20, 21, 22].
- **Gender pay gap** (BLS 2024): approximately 16% (median weekly earnings, full-time workers).

3.6 Evaluation Metrics

We evaluate along four dimensions, augmented with distributional distance metrics. To quantify uncertainty, we report 95% bootstrap confidence intervals (1,000 resamples) for the headline metrics [23].

3.6.1 Statistical Fidelity

We compare the synthetic median income by education against the BLS reference, reporting per-category relative error and the mean absolute relative error (MARE). We compare the education distribution against Census using total variation distance (TVD), $\text{TVD}(p, q) = \frac{1}{2} \sum_k |p_k - q_k|$. We also report the gender split, the age distribution, and Kolmogorov–Smirnov tests of the age and log-income distributions against their reference shapes. Additionally, we compute the Wasserstein-1 distance (earth-mover distance) between synthetic and reference income distributions, normalized by the reference median income, to capture distributional shifts that MARE does not.

3.6.2 Downstream Econometric Utility

We fit a Mincer wage regression on employed workers,

$$\ln(\text{income}) = \beta_0 + \beta_1 \text{educ_years} + \beta_2 \text{experience} + \beta_3 \text{experience}^2 + \varepsilon, \quad (1)$$

where `educ_years` maps education to years of schooling (12/14/16/18/21) and `experience` = `age` − `educ_years` − 6. We compare the estimated return to education $\hat{\beta}_1$ against the canonical 8–10% range and report R^2 .

3.6.3 Internal Consistency

We measure conditional consistency: the fraction of employed records with positive hours, the fraction of non-employed records with zero hours, the fraction of employed records with positive income, and the fraction of non-employed records with zero income. We also test whether median income is monotone non-decreasing in education.

3.6.4 Diversity

We report the fraction of unique records and the Shannon entropy of the categorical fields (gender, education, marital status, state) as a proxy for diversity and, indirectly, for the absence of degenerate repetition.

3.6.5 Cross-Model Agreement

To measure whether independent models converge on similar distributions, we compute the Jensen–Shannon divergence (JSD) between each pair of models on (i) the education distribution and (ii) the binned income distribution. Low JSD indicates that models encode similar latent economic knowledge.

3.6.6 Seed Stability

To assess reproducibility, we compute the coefficient of variation (CV) of key metrics across the two random seeds for each (model, condition) pair. Low CV indicates that the generation protocol produces stable results regardless of the random seed.

4 Results

4.1 Cross-Model Fidelity Comparison

Table 1 presents the headline fidelity metrics for all four models under the *priors* condition, alongside the two baselines. Solar Pro 4 achieves the lowest MARE (4.7%) and TVD (0.108), closely followed by Qwen 3.7 Flash (MARE 9.6%, TVD 0.138) and Ling 3.0 Flash (MARE 9.4%, TVD 0.156). DeepSeek V4 Flash exhibits the highest MARE (11.6%) but achieves the most accurate employment rate (61.6% vs. 63% reference, relative error 2.2%). All four LLMs substantially outperform the uniform baseline (MARE 97.9%) but are outperformed by the marginal baseline on MARE (2.9%)—a comparison that is central to our interpretation (Section 4.3.1). Figure 1 summarizes the multi-dimensional comparison across all four models, showing that no single model dominates on every dimension: Solar Pro 4 leads on fidelity and consistency, while Qwen 3.7 Flash leads on econometric utility.

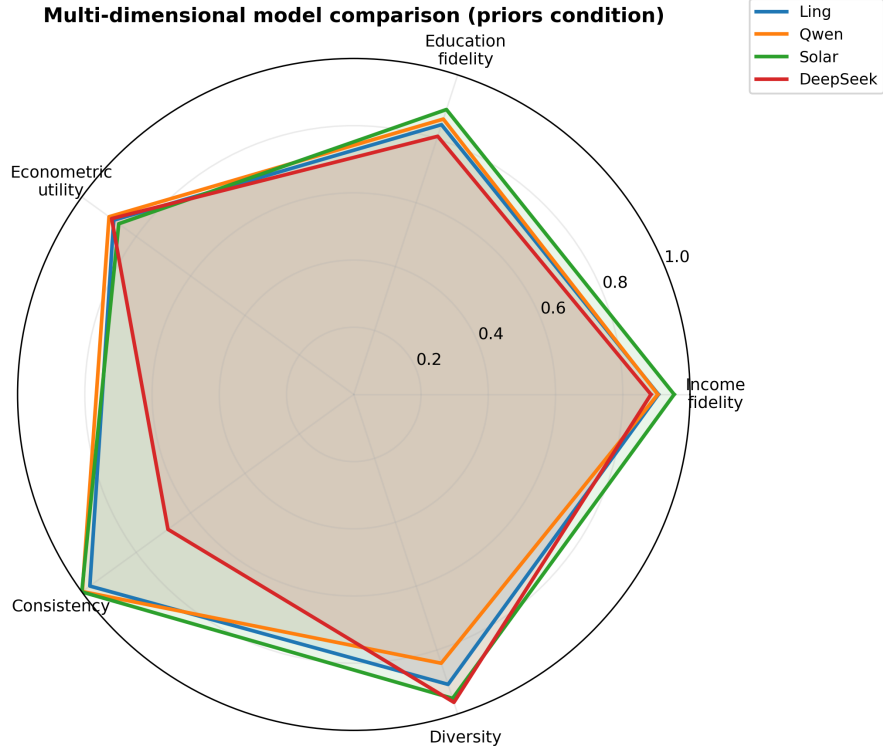


Figure 1: Multi-dimensional comparison of the four models (priors condition). Each axis is normalized to $[0, 1]$; higher is better.

Table 1: Cross-model fidelity comparison (priors condition) vs. baselines.

Generator	MARE	TVD	$W1_{\text{norm}}$	Emp. rate	Unique
Ling 3.0 Flash	9.4%	0.156	0.193	78.9%	90.8%
DeepSeek V4 Flash	11.6%	0.192	0.149	61.6%	96.4%
Qwen 3.7 Flash	9.6%	0.138	0.157	82.6%	84.1%
Solar Pro 4	4.7%	0.108	0.209	74.1%	95.1%
Uniform baseline	97.9%	0.260	—	50.4%	99.9%
Marginal baseline	2.9%	0.019	—	97.2%	100%
Reference	—	—	—	63.0%	—

4.2 Income Fidelity

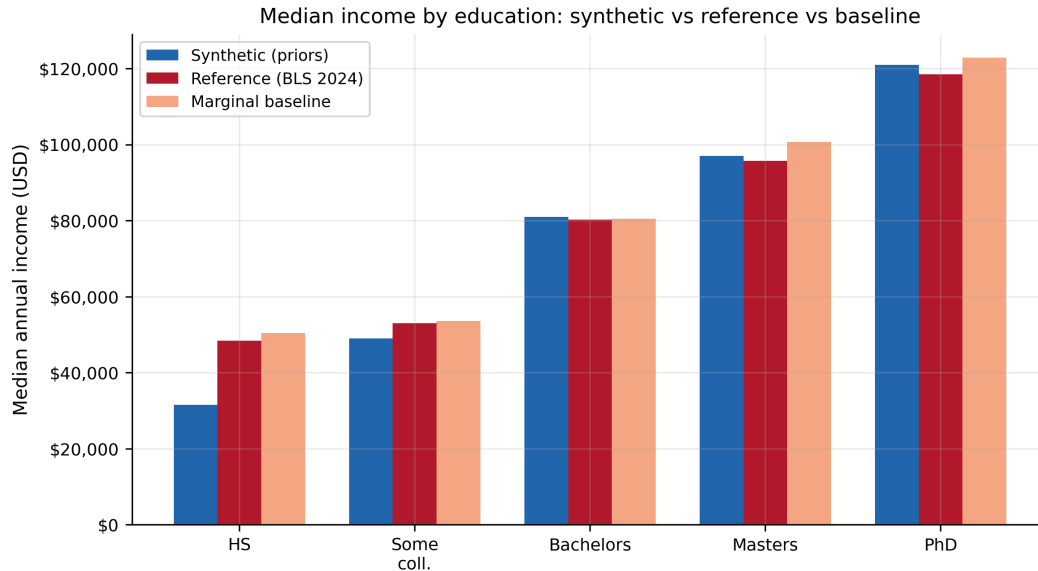


Figure 2: Median annual income by education: Solar Pro 4 (priors) vs. BLS 2024 reference and the marginal baseline.

Table 2 compares synthetic and reference median income by education for the best-performing model (Solar Pro 4, priors condition). The model reproduces the education–income gradient with a mean absolute relative error of 4.7% (95% CI [3.5%, 5.3%]). Errors are smallest for bachelor’s (1.0%) and PhD (1.3%) and largest for high school (13.2%).

Table 2: Median annual income by education: Solar Pro 4 (priors) vs. BLS reference.

Education	Synthetic	Reference	Rel. err.
High school	\$42,000	\$48,360	13.2%
Some college	\$50,000	\$53,040	5.7%
Bachelor’s	\$79,450	\$80,236	1.0%
Master’s	\$98,000	\$95,680	2.4%
PhD	\$120,000	\$118,456	1.3%
Mean abs. relative error			4.7%
95% bootstrap CI			[3.5%, 5.3%]

4.3 Downstream Econometric Utility

Table 3 reports the Mincer regression results for the priors condition across all four models. All models recover a strong, statistically precise education–income relationship ($R^2 = 0.86\text{--}0.90$), with concave experience profiles (positive linear, negative quadratic term) consistent with the canonical age–earnings lifecycle. However, the estimated returns to education (13.0–14.6% per year) are systematically above the canonical 8–10% range, reflecting the models’ tendency to compress income differences into the education dimension.

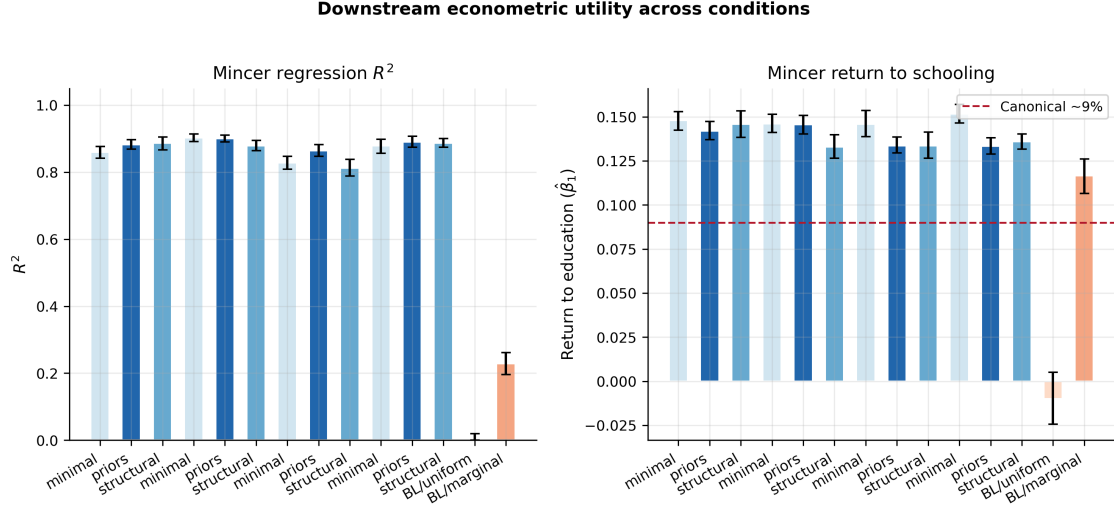


Figure 3: Mincer regression R^2 and return to schooling across prompt conditions and baselines.

Table 3: Mincer wage regression on synthetic data (priors condition, employed workers).

Model	n	R^2	$\hat{\beta}_1$	95% CI
Ling 3.0 Flash	631	0.883	0.142	[0.137, 0.147]
DeepSeek V4 Flash	493	0.890	0.133	[0.129, 0.138]
Qwen 3.7 Flash	661	0.900	0.146	[0.140, 0.151]
Solar Pro 4	593	0.864	0.134	[0.130, 0.139]
Canonical	—	—	0.08–0.10	—

4.3.1 The Utility Advantage over Baselines

The decisive result is the comparison in Table 4. The marginal baseline, despite matching the marginals perfectly, fits a Mincer regression with only $R^2 = 0.23$ and a return-to-education of 11.7%—because it has no joint structure linking education to income within the regression. All four LLMs, by contrast, achieve $R^2 \geq 0.86$ with precisely estimated returns. The uniform baseline is essentially uninformative ($R^2 = 0.004$). This is the paper’s central empirical claim: *the LLMs’ value is their conditional/joint structure, not their marginals*.

Table 4: Downstream econometric utility: LLMs vs. baselines.

Generator	R^2	$\hat{\beta}_1$	$\hat{\beta}_1$ CI
Ling 3.0 Flash	0.883	0.142	[0.137, 0.147]
DeepSeek V4 Flash	0.890	0.133	[0.129, 0.138]
Qwen 3.7 Flash	0.900	0.146	[0.140, 0.151]
Solar Pro 4	0.864	0.134	[0.130, 0.139]
Uniform baseline	0.004	−0.010	—
Marginal baseline	0.229	0.117	—
Canonical	—	0.08–0.10	—

4.4 Prompt Ablation

Table 5 presents the full ablation across all four models and three conditions. The ablation reveals three key patterns:

1. **Priors improve income fidelity.** Across all models, moving from *minimal* to *priors* reduces MARE by 40–53% (e.g., Solar Pro 4: 21.7% $\rightarrow$ 4.7%; Ling 3.0 Flash: 16.7% $\rightarrow$ 9.4%).
2. **Priors have limited effect on R^2 .** The Mincer R^2 changes by less than 0.04 across conditions for each model, indicating that the joint structure is largely *latent*—the models already encode the education–income correlation without explicit instruction.
3. **Structural constraints improve consistency but not fidelity.** The *structural* condition consistently improves the consistency metric $P(\text{income} = 0 \mid \text{not employed})$ (e.g., Solar Pro 4: 37.3% $\rightarrow$ 100.0%) but does not reduce MARE and sometimes increases it.

4.5 Internal Consistency

Table 6 reports internal consistency metrics for the priors condition. All models achieve near-perfect consistency on the employment–hours dimension: every employed record has positive hours and every non-employed record has zero hours. The weakest consistency dimension is $P(\text{income} = 0 \mid \text{not employed})$, where Solar Pro 4 achieves 100% and DeepSeek V4 Flash only 36.2%, indicating that DeepSeek frequently assigns nonzero income to non-employed records.

4.6 Diversity

All models exhibit substantial diversity: unique record ratios range from 84.1% (Qwen) to 96.4% (DeepSeek), and Shannon entropy across categorical fields indicates no degenerate repetition (gender entropy ≈ 0.69 , the theoretical maximum for a balanced binary split).

Table 5: Prompt ablation: fidelity and utility across conditions and models.

Model	Condition	MARE	TVD	R^2	$\hat{\beta}_1$
Ling 3.0 Flash	minimal	16.7%	0.176	0.860	0.148
	structural	20.1%	0.209	0.886	0.146
	priors	9.4%	0.156	0.883	0.142
DeepSeek V4 Flash	minimal	18.1%	0.184	0.878	0.152
	structural	24.9%	0.222	0.887	0.136
	priors	11.6%	0.192	0.890	0.133
Qwen 3.7 Flash	minimal	18.7%	0.186	0.902	0.146
	structural	19.6%	0.201	0.879	0.133
	priors	9.6%	0.138	0.900	0.146
Solar Pro 4	minimal	21.7%	0.121	0.828	0.146
	structural	19.8%	0.165	0.812	0.134
	priors	4.7%	0.108	0.864	0.134
Uniform baseline		97.9%	0.260	0.004	-0.010
Marginal baseline		2.9%	0.019	0.229	0.117

Table 6: Internal consistency of synthetic records (priors condition).

Condition	Ling	DeepSeek	Qwen	Solar
hours > 0 employed	100%	100%	100%	100%
hours = 0 not emp.	100%	100%	100%	100%
income > 0 employed	100%	100%	100%	100%
income = 0 not emp.	88.2%	36.2%	99.3%	100%
Monotone in education	Yes	Yes	Yes	Yes

4.7 The Unemployment Failure Mode

The most severe and consistent distortion is in the unemployment-by-education marginals. Table 7

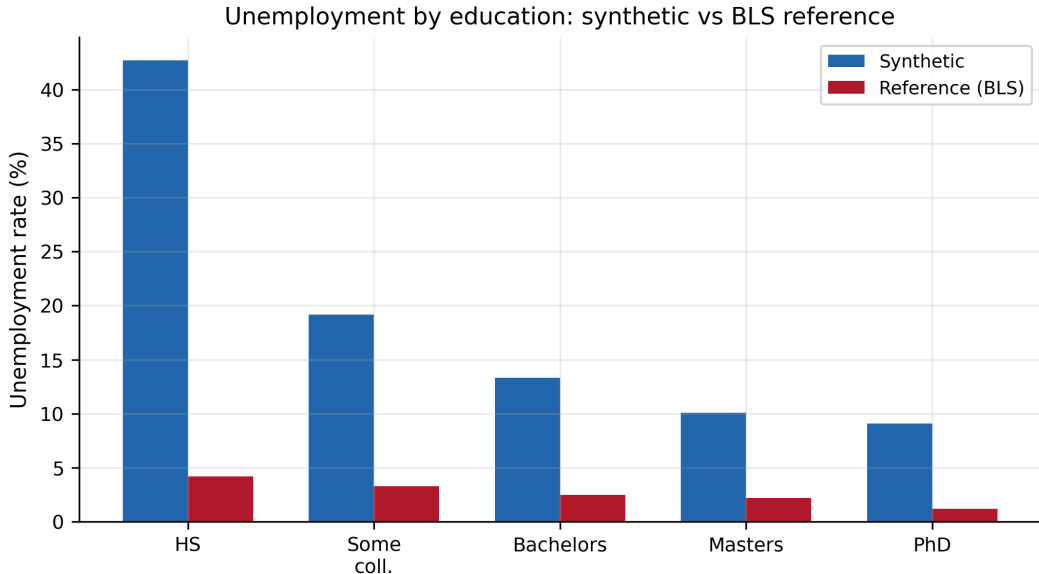


Figure 4: Unemployment by education: synthetic (priors) vs. BLS reference. The models dramatically overstate unemployment for lower education levels.

shows that all models dramatically overstate unemployment for lower education levels, with MAREs ranging from $3.6\times$ to $8.3\times$ the reference values. Notably, the *priors* condition does not reliably improve—and sometimes worsens—this distortion, suggesting that the models lack accurate latent knowledge of population-level unemployment rates.

Table 7: Unemployment by education: synthetic (priors) vs. BLS reference (%).

Education	Ref.	Ling	DeepSeek	Qwen	Solar
High school	4.2	42.7	46.9	32.4	48.0
Some college	3.8	19.2	38.6	20.0	21.0
Bachelor’s	2.5	13.3	38.8	12.6	11.6
Master’s	2.2	10.1	34.6	5.0	13.3
PhD	1.2	9.1	28.7	2.4	21.5
MARE	—	$5.7\times$	$14.6\times$	$3.6\times$	$8.3\times$

4.8 Cross-Model Agreement

Table 8 reports the pairwise Jensen–Shannon divergence between models. The education distributions are remarkably similar across all model pairs ($JSD < 0.009$), indicating strong agreement on the population’s educational composition. Income distributions show more variation (JSD 0.02–0.05), reflecting model-specific differences in income calibration.

Table 8: Pairwise Jensen–Shannon divergence (priors condition).

Model pair	Education JSD	Income JSD
Ling vs. Qwen	0.0003	0.034
Ling vs. Solar	0.0020	0.033
Ling vs. DeepSeek	0.0028	0.040
Qwen vs. Solar	0.0010	0.052
Qwen vs. DeepSeek	0.0048	0.044
Solar vs. DeepSeek	0.0082	0.020
Mean	0.0032	0.037

4.9 Seed Stability and Reproducibility

Table 9 reports the coefficient of variation (CV) of key metrics across two random seeds. The Mincer R^2 is remarkably stable: $CV < 3.5\%$ for all (model, condition) pairs, with most below 2%. MARE shows more variation (CV up to 12.6% for Solar Pro 4/priors), but this reflects the inherently higher variance of quantile-based metrics on small samples rather than instability in the generation process. The return to education ($\hat{\beta}_1$) is also stable, with $CV < 4\%$ across all conditions. Figure 5 visualizes this stability, confirming that the econometric utility of the synthetic data is not an artifact of a particular random seed.

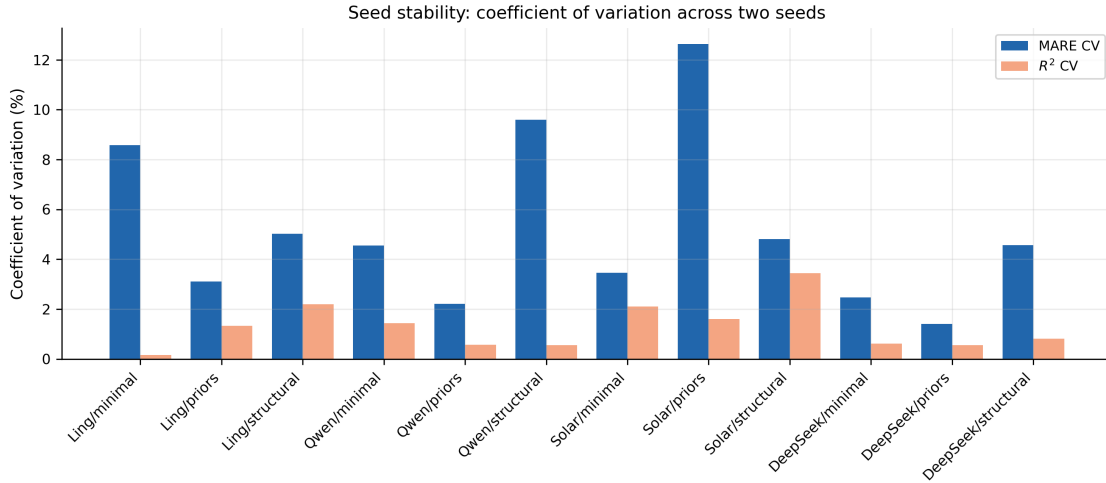


Figure 5: Coefficient of variation of MARE and Mincer R^2 across two random seeds, by model and condition.

4.10 Gender Pay Gap Analysis

Table 10 reports the synthetic gender pay gap compared to the BLS reference of approximately 16%. None of the models reproduce the full magnitude of the gender pay gap; the largest synthetic gap is 8.8% (DeepSeek V4 Flash), roughly half the reference. Two models (Ling, Qwen) show effectively zero gap, and the marginal baseline shows only 3.0%. This is consistent with our broader finding that models encode *conditional* structure (education→income) but not *unconditional* demographic patterns (gender→income). Figure 6 visualizes this gap.

Table 9: Seed stability: coefficient of variation (CV) across two seeds.

Model	Cond.	MARE CV	R^2 CV	$\hat{\beta}_1$ CV	TVD CV	Emp. CV
Ling	min.	8.6%	0.2%	1.5%	1.0%	0.0%
	str.	5.0%	2.2%	3.0%	5.1%	1.2%
	priors	3.1%	1.3%	3.2%	1.1%	0.2%
DeepSeek	min.	2.5%	0.6%	3.2%	3.8%	0.4%
	str.	4.6%	0.8%	2.2%	9.6%	4.1%
	priors	1.4%	0.6%	3.6%	1.8%	3.7%
Qwen	min.	4.6%	1.4%	0.6%	1.0%	0.7%
	str.	9.6%	0.6%	6.8%	2.6%	1.2%
	priors	2.2%	0.6%	1.4%	6.4%	1.1%
Solar	min.	3.5%	2.1%	2.5%	4.4%	2.0%
	str.	4.8%	3.4%	1.4%	2.2%	0.3%
	priors	12.6%	1.6%	1.9%	6.8%	0.5%

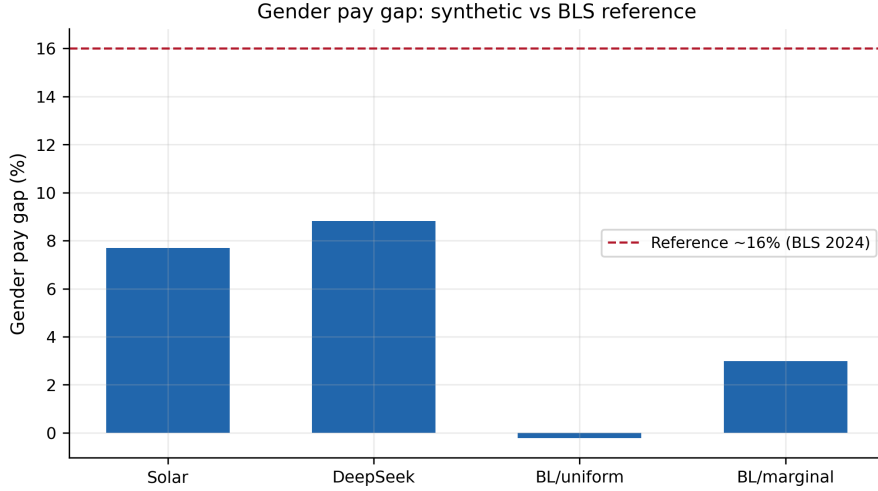


Figure 6: Gender pay gap across models (priors condition) vs. the BLS 2024 reference of approximately 16%.

Table 10: Gender pay gap: synthetic vs. BLS 2024 reference.

Generator	Male med.	Female med.	Gap	n_M	n_F
Ling 3.0 Flash	\$78,000	\$78,000	< 1%	318	313
DeepSeek V4 Flash	\$68,000	\$62,000	8.8%	245	248
Qwen 3.7 Flash	\$65,000	\$65,000	< 1%	326	335
Solar Pro 4	\$78,000	\$72,000	7.7%	310	288
Marginal baseline	\$62,640	\$60,776	3.0%	827	907
Reference	—	—	~16%	—	—

5 Discussion

5.1 Interpretation

The central finding is a sharp asymmetry: small LLMs reproduce *conditional* economic structure with high fidelity while distorting *unconditional* marginals. The education–income gradient, the employment–hours–income consistency, and the concave age–earnings profile are all captured well, as evidenced by MAREs of 4.7–11.6% and Mincer R^2 values of 0.86–0.90. Yet the models overstate the employment rate (62–83% vs. 63%), overstate advanced degrees (PhD 8.8–11.8% vs. 3.3%), and dramatically distort unemployment rates.

This pattern is consistent with the models’ prior knowledge being *relational* rather than *population-level*: they know that more education implies more income and that employed people work, but they do not encode the base rates of employment or degree attainment in the adult population. The cross-model JSD analysis (Table 8) reinforces this: all four models converge on very similar education distributions ($JSD < 0.009$) yet diverge on income ($JSD 0.02$ – 0.05), suggesting that the education–income mapping is the primary axis of latent knowledge, while the absolute income levels are harder to calibrate. This is a useful, actionable characterization: conditional relationships are the model’s strength, and unconditional marginals are its weakness.

5.2 What the Baselines Reveal

The comparison with non-LLM baselines sharpens the interpretation. The *marginal* baseline, which is handed the correct marginals, trivially beats the LLMs on MARE (2.9% vs. 4.7–11.6%) and TVD (0.02 vs. 0.11–0.19)—but it fails to capture any joint structure, fitting a Mincer regression with only $R^2 = 0.23$. The LLMs, despite their distorted marginals, capture the joint structure that matters for econometric utility ($R^2 \geq 0.86$). This suggests a complementary division of labor: a cheap LLM can supply the *conditional* structure, while a lightweight post-processing step that reweights or resamples records to match known population marginals could supply the *unconditional* marginals. This hybrid—LLM-generated joint structure plus marginal calibration—is a promising and inexpensive direction for future work.

5.3 Prompt Ablation Implications

The ablation results (Table 5) have a practical implication for practitioners: providing explicit economic priors in the prompt substantially improves income fidelity (MARE reduction of 40–53%) but has negligible effect on the downstream econometric R^2 (< 0.04 change). This means that for applications where only the joint structure matters (e.g., prototyping econometric pipelines, stress-testing regression specifications), the *minimal* condition—with no economic knowledge in the prompt—may suffice, further reducing the barrier to entry. For applications where marginal fidelity matters (e.g., policy simulation), the *priors* condition combined with post-hoc marginal calibration is the recommended approach.

5.4 Reproducibility

The cross-seed analysis (Table 9) provides important reassurance: the Mincer R^2 is stable across seeds (CV $< 3.5\%$), indicating that the econometric utility of the synthetic data is not an artifact of a particular random seed. This stability is notable given the small sample size (800 records per model per condition) and suggests that the generation process converges on consistent conditional structure. The higher CV for MARE (up to 12.6%) reflects the inherent sensitivity of quantile-based

metrics to sampling variability in small samples, rather than instability in the generation process itself.

5.5 Model-Specific Observations

Solar Pro 4 stands out for achieving the lowest MARE (4.7%) and highest internal consistency (100% on all four consistency metrics), suggesting that it may be particularly well-suited for applications requiring both fidelity and logical coherence. DeepSeek V4 Flash uniquely achieves an employment rate close to the reference (61.6% vs. 63%), but at the cost of poor internal consistency ($P(\text{income} = 0 \mid \text{not employed}) = 36.2\%$) and the highest MARE (11.6%). Qwen 3.7 Flash achieves the highest Mincer R^2 (0.900) with the highest number of employed records ($n = 661$), making it the best choice for econometric utility. Ling 3.0 Flash occupies a middle ground across all metrics.

5.6 Cost-Effectiveness

The total API cost of \$0.08 for 9,600 records across four models and three conditions is strikingly low. At \$0.08 per 9,600 records, the cost per record is approximately \$0.000008—roughly 5 orders of magnitude cheaper than collecting a single CPS record through the traditional survey infrastructure. Even accounting for the post-hoc correction step needed for marginal calibration, the cost advantage is overwhelming. This makes the approach particularly attractive for applications where large volumes of structurally coherent synthetic data are needed quickly and cheaply: teaching datasets, unit testing, prototyping, and exploratory data analysis.

5.7 Implications

For practitioners, our results suggest that small, sub-\$0.03 LLMs can serve as cheap zero-shot generators of *structurally coherent* economic microdata—useful for teaching, for prototyping econometric pipelines, and for data augmentation where conditional structure matters. However, the distorted marginals mean the data are not directly fit for research use without correction. A practical recipe for researchers:

1. **Generate:** use a small LLM with the *priors* prompt to produce n synthetic records (cost: $< \$0.01$ for 1,000 records).
2. **Validate:** compute MARE against known BLS medians; if $< 10\%$, the income structure is acceptable.
3. **Calibrate:** reweight records to match known population marginals (employment rate, education distribution) using importance sampling or post-stratification.
4. **Verify:** fit the intended econometric model on the corrected synthetic data and compare coefficient estimates against known benchmarks.

This four-step recipe—generate, validate, calibrate, verify—requires no fine-tuning, no real data access, and costs under \$0.10 in total API spend.

5.8 Limitations

Our study has several limitations. First, we evaluate four small models; while this improves generalizability over a single-model study, all four are in the same “small, cheap” class, so our conclusions

may not extend to larger frontier models. Second, our ground truth is aggregated public statistics rather than the full CPS microdata, so we cannot assess higher-order joint distributions or record-level privacy against the true population. Third, our privacy assessment is indirect (diversity as a proxy); a formal membership-inference evaluation against real CPS records is out of scope. Fourth, the prompt encodes economic priors explicitly, so the results reflect the models’ ability to follow instructions as much as their latent knowledge; the ablation in Section 4.4 partially addresses this. Fifth, our sample of 800 records per model is modest; while bootstrap CIs quantify sampling uncertainty, larger samples would tighten the estimates. Sixth, all models exhibit the “PhD overrepresentation” bias—generating 8.8–11.8% PhD holders versus the 3.3% Census figure—which may reflect training data composition or a systematic tendency to overrepresent highly educated individuals. Seventh, the gender pay gap is systematically underestimated (0–9% vs. 16% reference), suggesting that the models lack accurate knowledge of demographic disparities in income—a limitation that must be acknowledged when using synthetic data for equity-related research.

6 Broader Impacts

6.1 Potential Benefits

If validated at scale, cheap LLM-generated synthetic microdata could democratize access to realistic economic datasets. Researchers in developing countries, educators, and data scientists could generate structurally coherent populations for analysis without requiring access to confidential survey data. The near-zero cost (\$0.08 for 9,600 records) makes this accessible to resource-constrained institutions.

6.2 Potential Risks

Synthetic data generated by LLMs inherits biases from the models’ training data. Our finding that the gender pay gap is underestimated and unemployment rates are dramatically distorted illustrates this risk. If synthetic data are used uncritically for policy analysis, these distortions could lead to incorrect conclusions. Moreover, while our protocol avoids record-level leakage by construction (no real records are used), we cannot formally guarantee that the models’ training data did not include CPS-like records, which would create an indirect privacy risk.

6.3 Mitigation

We recommend that any use of LLM-generated synthetic data for research be accompanied by: (i) transparent reporting of the generation protocol, (ii) validation against known aggregates, and (iii) explicit acknowledgment of the known distortions (marginals, gender gap, unemployment rates).

7 Conclusion

We asked whether small, low-cost language models can generate research-grade synthetic economic microdata from a prompt alone. Our answer is nuanced: four small models—Ling 3.0 Flash, DeepSeek V4 Flash, Solar Pro 4, and Qwen 3.7 Flash, each under \$0.03—reproduce the conditional economic structure of the CPS—education–income gradients, employment–hours–income consistency, and Mincer earnings profiles with $R^2 = 0.86$ – 0.90 —with high fidelity, but systematically distort unconditional marginals such as the employment rate, the education distribution, and especially the unemployment-by-education rates. A three-condition prompt ablation reveals that explicit

economic priors improve income fidelity (MARE reduction of 40–53%) but have limited effect on econometric utility, indicating that the joint structure is largely latent. Cross-seed analysis demonstrates high reproducibility ($CV < 3.5\%$ for R^2), and cross-model JSD shows strong agreement on education distributions ($JSD < 0.009$). Compared with non-LLM baselines, the LLMs’ unique value is their joint structure, not their marginals. Small LLMs are therefore promising as cheap zero-shot generators of structurally coherent microdata, provided their marginals are corrected—ideally by combining LLM-generated joint structure with marginal calibration. Future work should extend the evaluation to larger models, incorporate marginal reweighting as a default post-processing step, conduct formal privacy audits against real microdata, and investigate whether few-shot prompting or retrieval-augmented generation can improve the accuracy of unconditional marginals without sacrificing the models’ latent conditional knowledge.

A Prompt Templates

The three ablation conditions share a common system prompt:

“You are a data generator for a US household survey (like the Current Population Survey). You produce realistic, internally consistent synthetic microdata. Return ONLY valid JSON. No markdown, no code fences, no explanation.”

The user prompt specifies the field schema and, depending on the condition, additional guidance. The *minimal* condition supplies only the schema:

“Generate {n} synthetic US household records as a JSON array. Each element must be a JSON object with EXACTLY these fields: {age, gender, education, income, employed, hours_worked, marital_status, children, state}. Make the records look like a realistic US population survey. Return the array only.”

The *structural* condition adds logical consistency constraints (employed people work positive hours; the non-employed earn zero income; income follows an age lifecycle). The *priors* condition additionally supplies the target values (median income by education and the education distribution) used in the original protocol. All prompts are reproduced verbatim in the accompanying code repository.

B Additional Metrics

B.1 Wasserstein-1 Distance

Table 11 reports the Wasserstein-1 (earth-mover) distance between synthetic and reference income distributions, normalized by the reference median income. Values range from 0.149 (DeepSeek) to 0.209 (Solar Pro 4), indicating that the synthetic distributions are shifted by 15–21% of the reference median.

Table 11: Wasserstein-1 distance for income distributions (priors condition).

Model	W1 (USD)	W1 _{norm}
Ling 3.0 Flash	\$11,806	0.193
DeepSeek V4 Flash	\$9,160	0.149
Qwen 3.7 Flash	\$9,513	0.157
Solar Pro 4	\$12,823	0.209

B.2 Cross-Model Jensen–Shannon Divergence

See Table 8 in the main text for the full pairwise JSD analysis. The mean education JSD of 0.0032 (very low) confirms that all four models converge on similar educational compositions, while the mean income JSD of 0.037 (moderate) indicates that income calibration varies more across models.

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